QUANTITATIVE ANALYSIS OF BIODIVERSITY IN HAVELI DISTRICT, AJK, PAKISTAN

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Objectives of the Study

1. Identify species in Haveli District habitats
2. Measure biodiversity and relative abundance
3. Evaluate species richness and conservation status
4. Explore unstudied areas

INTRODUCTION



Wildlife Diversity in Haveli District, AJK

- **Location:** Haveli District is part of Azad Jammu and Kashmir, in Poonch Division with an area of 598 km2.
- **Geography:** High altitude, approx. 8,000 feet above sea level, heavy snowfall throughout the year.
- **Wildlife:** Diverse species due to varied topography and vegetation.
- **Protected Areas:** Khewra Wildlife Sanctuary, Manglot Wildlife Park, crucial for wildlife conservation.
- **Tourist Attractions:** Well-known tourist locations like Darra Haji Peer, Lasdana, and Neel Kunth.
- 12 species of reptiles, 202 bird species and 30 species of mammals were recorded in Tolipir National Park.

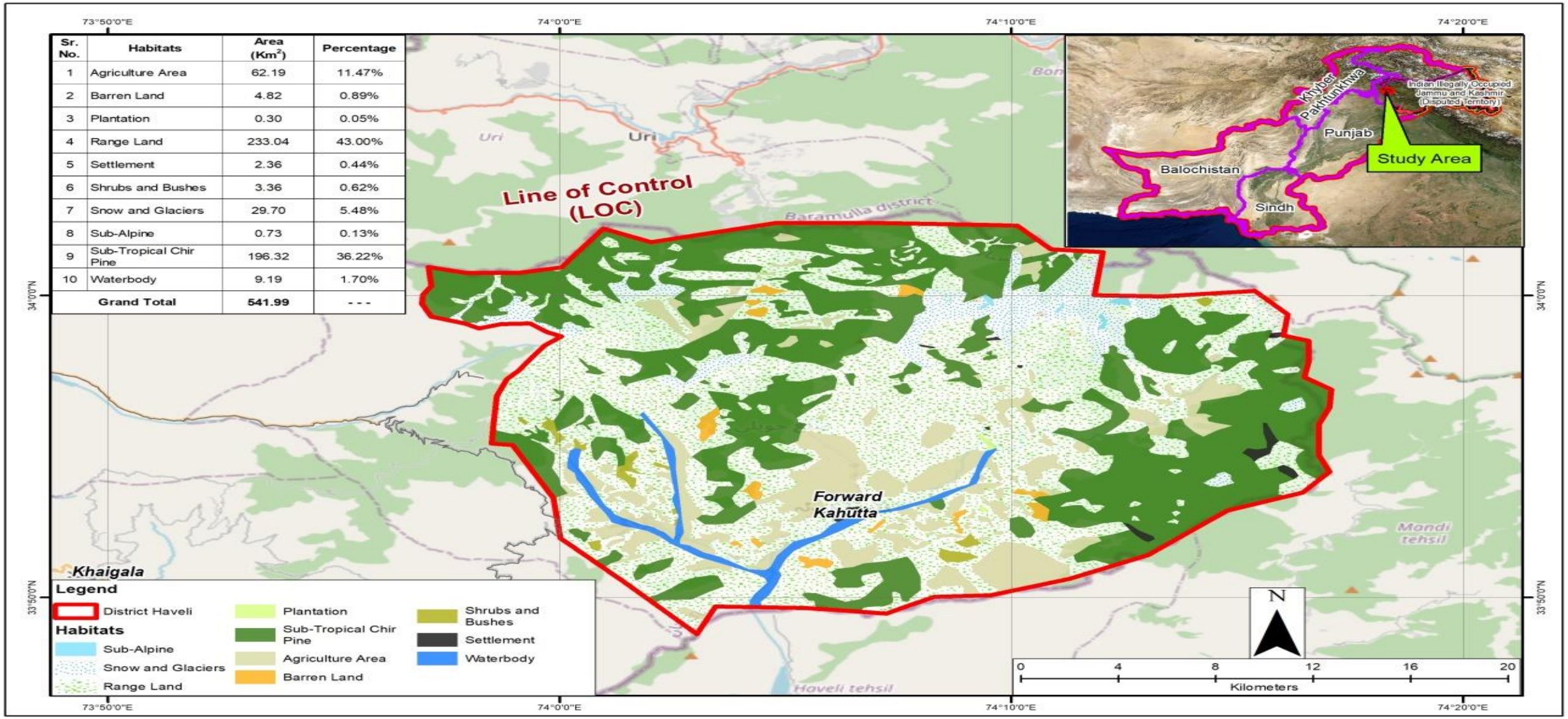
Gaps in the Literature

- Lack of Research and Documentation
- Importance of Comprehensive Study
- Revealing Biodiversity Insights
- Contribution to Understanding
- Informed Conservation Strategies

MATERIALS AND METHODS



Nine surveys were conducted in District Haveli AJK from September 2022 to June 2023. The District is home to ten different habitat types; Agriculture Area, Barren Land, Plantation , Range Land, Settlement, Shrubs and Bushes, Snow and Glaciers, Sub-Alpine Sub-Tropical Chir Pine, and Waterbody



Sr. No.	Habitats	Area (Km^2)	Percentage
1	Agriculture Area	62.19	11.47%
2	Barren Land	4.82	0.89%
3	Plantation	0.30	0.05%
4	Range Land	233.04	43.00%
5	Settlement	2.36	0.44%
6	Shrubs and Bushes	3.36	0.62%
7	Snow and Glaciers	29.70	5.48%
8	Sub-Alpine	0.73	0.13%
9	Sub-Tropical Chir Pine	196.32	36.22%
10	Waterbody	9.19	1.70%
Grand Total		**541.99**	- - -

Study Map of Haveli District

Equipments

- Garmin GPS map 76CSx,
- Binoculars (Bushnell power view, 60 X 90 m,
- Harrier 65mm ED Spotting Scope,
- Range finder and camera (Nikon p-900).
- Field Guides including Roberts, Khan and Mirza, Grimmet et al., and Mirza and Wasiq.

Survey Methods and Techniques

1. **Comprehensive Survey Coverage:**

- Monthly Surveys
- GIS Implementation

2. **Standardized Survey Techniques:**

- Direct Method.
- Bird Identification:

  On-spot Observation, Photographs, Field Guides

3. Point Count was used as a **survey method** for counting birds at specific locations.

Data Analysis

- Relative Abundance: $R.A = n/N(100)$
- Shannon Wiener Index: $H = -\sum[(pi) \times \ln(pi)]$
- Simpsons Index: $D = \sum n(n-1)/ N(N-1)$
- Species evenness $= H'/\ln(S)$
- Sorensen index $= 2C / (A + B)$
- Jaccard index: $J(A, B) = |A \cap B| / |A \cup B|$

RESULTS

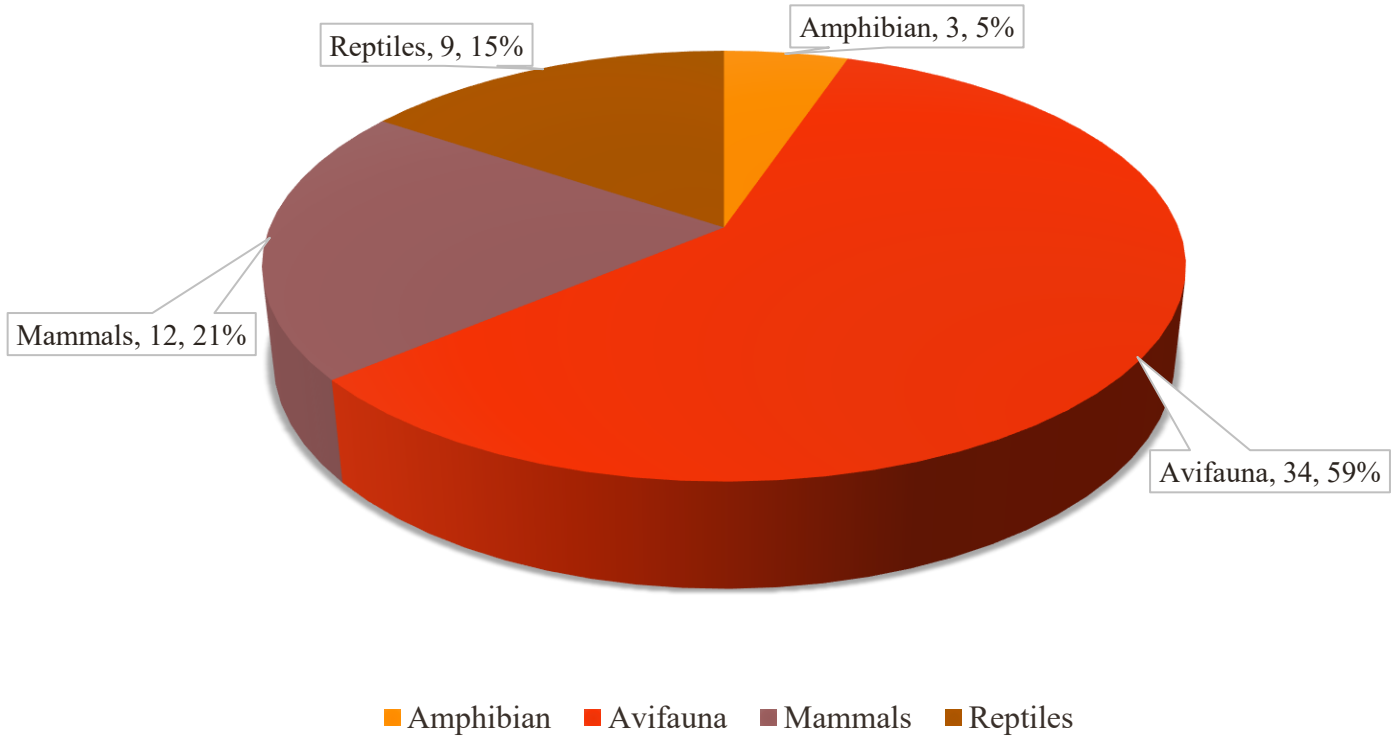


During the whole study duration a total of 493 individuals of 58 species of vertebrates (birds, reptiles, mammals and amphibians) that belongs to 12 orders and 32 families were observed.

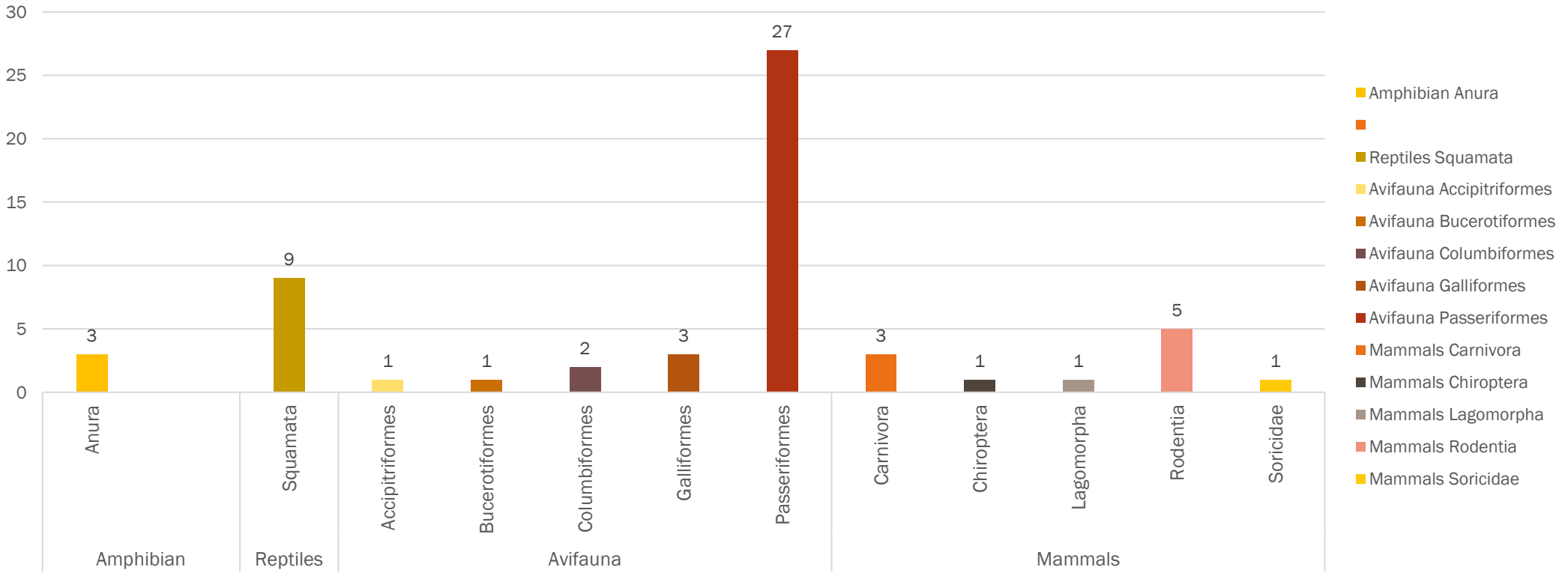
Individuals Abundance in Various Taxonomic Groups

Species Type	Number of Individuals
Amphibian	16
Reptiles	62
Avifauna	332
Mammals	83
Total	493

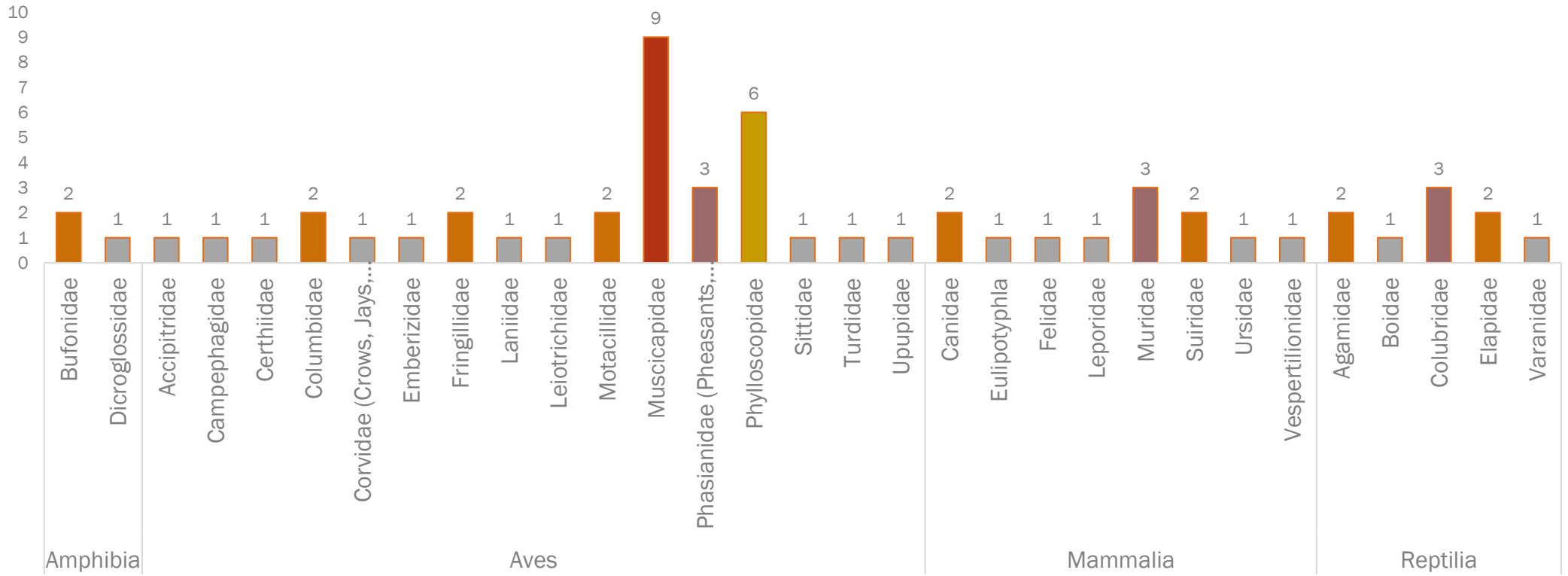
Species Distribution According to Class



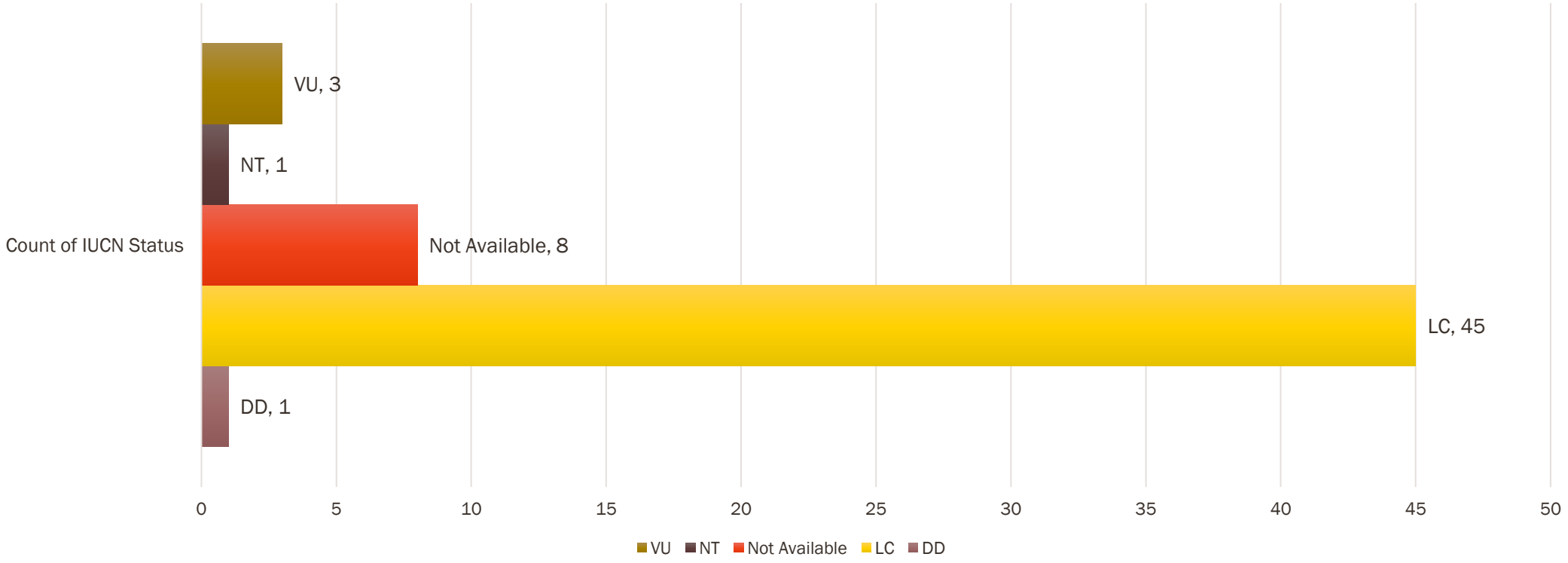
Species Distribution Based on Taxanomic Orders



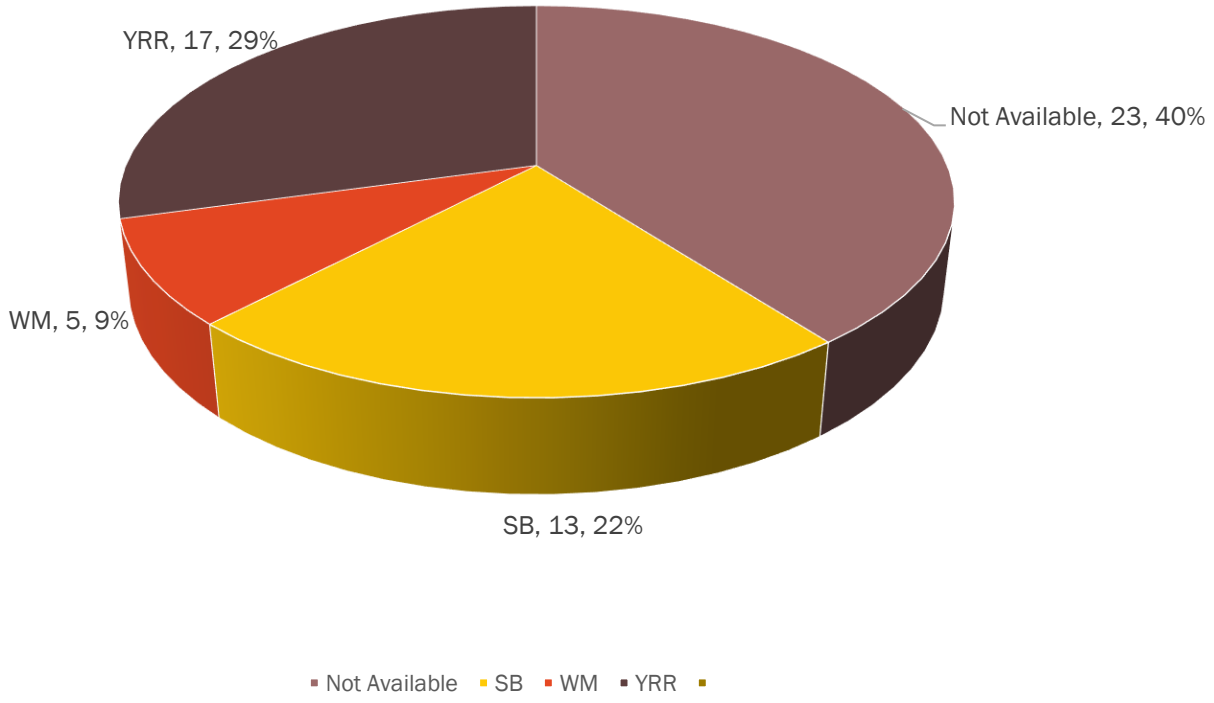
Species Distribution Based on Families



IUCN Trend of Species



Species Abundance Based on Occurrence

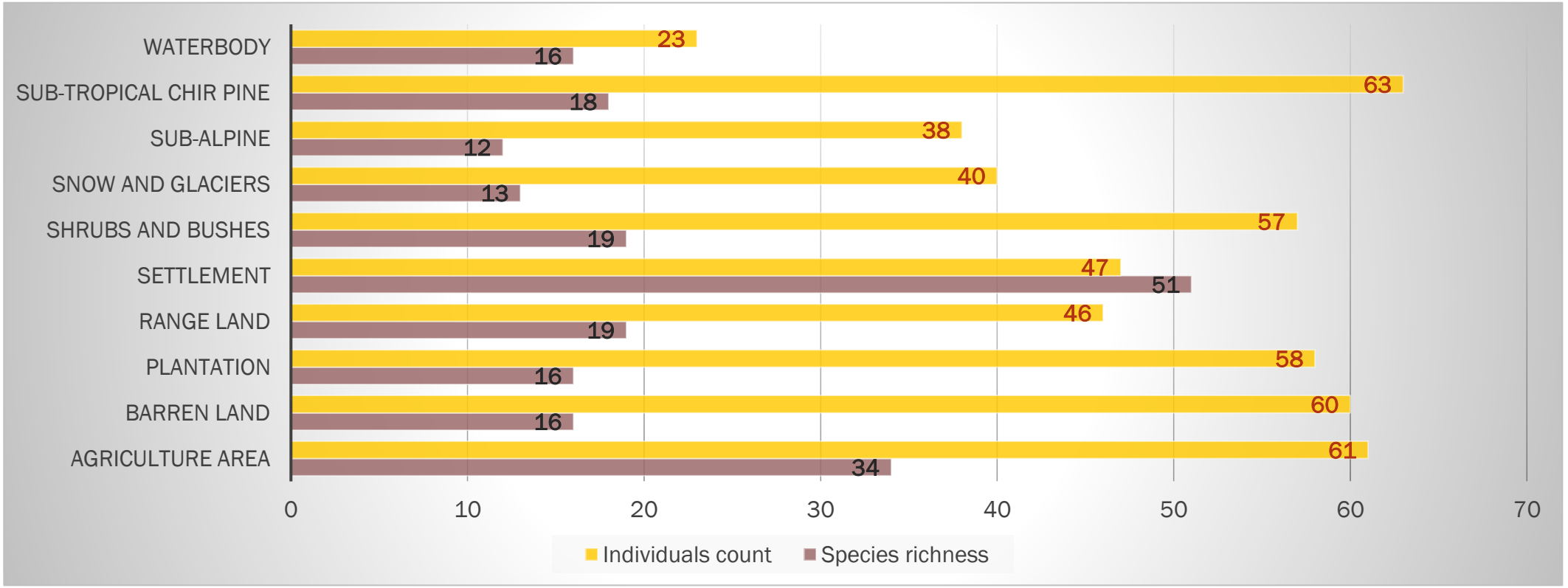


Species Richness and Abundance in Haveli District AJK

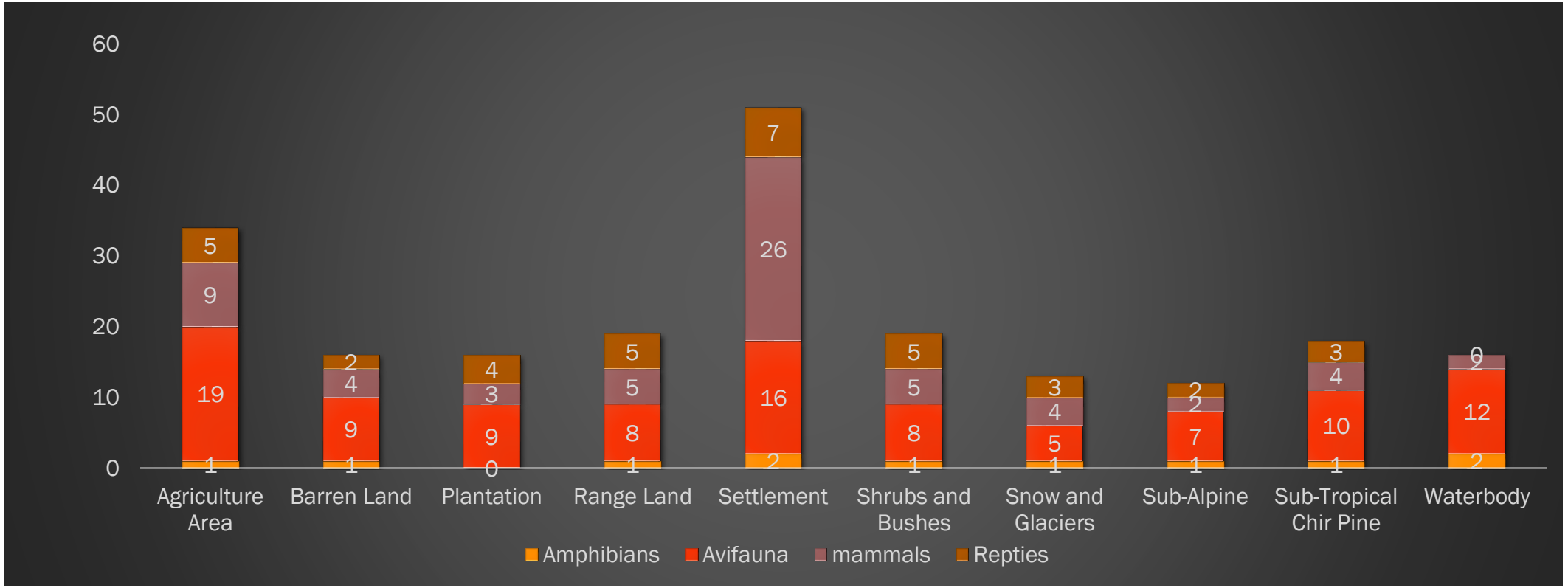
- The Shannon Weinner value was 3.818 and Simpson's value was 0.973 of the whole study area. The species evenness of the study area was 0.213.
- The most abundant species in different Classes are:

Class	Common Name	Scientific Name	R.A
Amphibians	Indus Valley Toad	Bufo stomaticus	1.62
Reptiles	Common Sand Boa	Eryx johnii	4.46
Avifauna	Common Hoopoe	Upupa epops	8.72
Mammals	Indian Giant Flying Squirrel	Petaurista petaurista	2.64
	Kashmir Fox	Vulpes vulpes griffithi	

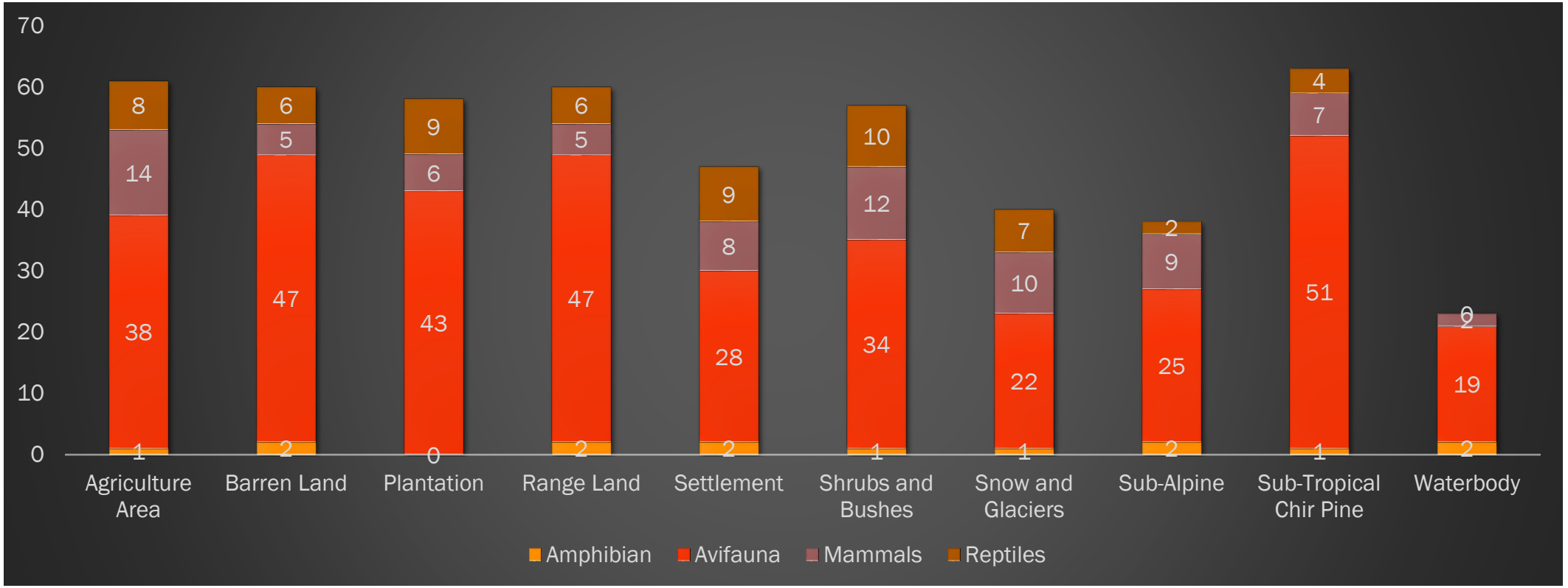
Species richness & Individuals count in different habitats



Species richness of Taxonomic Groups in Various Habitats



Individuals Abundance of Taxonomic Groups in Various Habitats



Diversity Indices of Different Habitats

Habitat	Species Richness	Shannon-Weiner	Simpsons Index 1-D	Species Eveness
Agriculture Area	34	3.36	0.97	0.28
Barren Land	16	2.98	**0.94**	0.34
Plantation	16	3.28	0.97	0.35
Range Land	19	3.26	0.97	0.33
Settlement	**51**	3.37	**0.98**	**0.25**
Shrubs and Bushes	19	**3.40**	0.97	0.33
Snow and Glaciers	13	3.03	0.96	0.38
Sub-Alpine	**12**	3.18	0.97	**0.39**
Sub-Tropical Chir Pine	18	2.98	0.96	0.33
Waterbody	16	**2.60**	0.95	0.34

The Most Abundant Species in Various Habitats

Habitats	Amphibians	Reptiles	Avifauna	Mammals
Agriculture Area	Burrowing Frog	Agrore Valley Agama	Common Hoopoe	Red Fox
Barren Land	Indus Valley Toad	Bengal Monitor	Bar-tailed Treecreeper	Himalayan Pipistrelle
Plantation	Absent	Common Sand Boa	White Wagtail	Asiatic Black Bear
Range Land	Burrowing Frog	Common Sand Boa	Black Redstart	Asiatic Pygmy Shrew
Settlement	Himalayan Toad	Common Sand Boa	Indian Peafowl	Small Kashmir Flying Squirrel
Shrubs and Bushes	Indus Valley Toad	Kashmir Cliff Racer	Cheer Pheasant	Asiatic Black Bear
Snow and Glaciers	Indus Valley Toad	Common Sand Boa	Lemon-rumped	Panther
Sub-Alpine	Himalayan Toad	Kashmir Cliff Racer	Common Hoopoe	Panther
Sub-Tropical Chir Pine	Himalayan Toad	Banded Kukri Snake	Common Hoopoe	Kashmir Fox
Waterbody	Indus Valley Toad	Absent	Common Hoopoe	House Mouse

Sorensen's Indices of Different Habitats

Habitats	Agriculture Area	Waterbody	Settlement	Range Land	Plantation	Barren Land	Shrubs and Bushes	Snow and Glaciers	Sub-Alpine	Sub-Tropical Chir Pine
Agriculture Area		0.333	0.518	0.485	0.554	0.429	0.491	0.356	0.444	0.58
Waterbody			0.342	0.347	0.246	0.385	**0.25**	0.317	0.459	0.348
Settlement				0.516	0.495	0.411	0.48	0.344	0.305	0.454
Range Land					0.442	0.358	0.543	0.465	0.452	0.366
Plantation						0.44	0.678	0.448	0.437	0.512
Barren Land							0.41	0.42	0.326	**0.617**
Shrubs and Bushes								0.474	0.442	0.516
Snow and Glaciers									0.461	0.427
Sub-Alpine										0.475
Sub-Tropical Chir Pine										

Jaccard Indices of all habitats

Habitats	Agriculture Area	Waterbody	Settlement	Range Land	Plantation	Barren Land	Shrubs and Bushes	Snow and Glaciers	Sub-Alpine	Sub-Tropical Chir Pine
Agriculture Area		0.2	0.35	0.32	0.383	0.273	0.325	0.216	0.285	0.409
Waterbody			0.206	0.21	0.14	0.238	0.142	0.188	0.297	0.211
Settlement				0.347	0.329	0.258	0.316	0.208	0.18	0.294
Range Land					0.283	0.218	0.373	0.303	0.292	0.224
Plantation						0.282	0.513	0.289	0.28	0.344
Barren Land							0.258	0.265	0.195	0.447
Shrubs and Bushes								0.31	0.283	0.348
Snow and Glaciers									0.3	0.271
Sub-Alpine										0.311
Sub-Tropical Chir Pine										

Conclusion



Conclusion

- A total of 58 species in which three amphibians, nine reptiles, 34 avifauna and 12 mammals were present belonging to 12 orders and 32 families.
- The conservation status of the species varied, with 45 listed as "Least Concern," 8 as "Not Available," and one species each as "Data Deficient," "Near Threatened," and three were "Vulnerable."
- The study area exhibited a high level of species diversity, as indicated by the Shannon-Weiner Diversity Index (H=3.81) and the Simpson's Diversity Index (D=0.973).
- The species evenness value of 0.212 suggested uneven distribution within the study area.
- Settlement was the most diverse habitat among all other habitats.
- The results from Sorenson Indicies revealed that the maximum similarity in between Barren Land and Sub-Tropical-Chir Pine while Jaccard indices, the maximum similarity was observed between Plantation and Shrubs and Bushes.

Threats

- Habitat Destruction
- Climate Change
- Overexploitation
- Human-Wildlife Conflict
- Lack of Awareness and Conservation Efforts

Recommendations

- Quantify the extent of habitat loss and fragmentation in Haveli using remote sensing and GIS tools.
- Gather data on conflicts arising from crop raiding, livestock predation, or property damage to understand its implications for wildlife conservation.
- Creating and managing protected areas to shield wildlife species from disturbances.
- Restoring damaged habitats through reforestation.
- Supporting endangered species conservation through preservation and captive breeding.
- Raising awareness among local communities through educational programs and community engagement.


thank
you